mapping technique, the Born–Markov master equation can then be immediately written down by substituting $\hat{X} \longrightarrow \hat{\sigma}_z$ in the general expression (5.22) for the master equation for quantum Brownian motion,

$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\rho}_{\mathcal{S}}(t) = -\mathrm{i}\big[\hat{H}_{\mathcal{S}}, \hat{\rho}_{\mathcal{S}}(t)\big] - \int_0^\infty \mathrm{d}\tau \, \{\nu(\tau)\,[\hat{\sigma}_z, [\hat{\sigma}_z(-\tau), \hat{\rho}_{\mathcal{S}}(t)]] - \mathrm{i}\eta(\tau)\,[\hat{\sigma}_z, \{\hat{\sigma}_z(-\tau), \hat{\rho}_{\mathcal{S}}(t)\}]\}\,. \quad (5.193)$$